VANDA RESEARCH

ENGINEERING · TAKE-HOME EXERCISE

Retail Equity Analytics Dashboard

A full-stack exercise that mirrors the work we actually do — from data pipeline to interactive chart.

24–48h effort   Backend: **Python**   Frontend: **TypeScript**   Data: **Twelvedata**

Thank you for taking the time to interview with us. This exercise is your opportunity to show how you work end to end — how you make decisions, weigh trade-offs, and turn an open-ended brief into a working product. A focused, coherent, well-explained slice will always score higher than a broad but unfinished one.

THE BRIEF

Build an **authenticated dashboard** that visualises **price and volume over time** for a small set of stock tickers, alongside **derived statistics** — period % change, high/low, and cumulative volume. Only authenticated users may retrieve data.

Behind it sits a simple data pipeline: a routine that fetches **daily OHLCV** data from the Twelvedata API (create your own free key) and stores it. Your app then serves data from your own store — it should **never call Twelvedata at request time**, and the key should never reach the browser.

HOW THIS WORKS

> This brief is **intentionally under-specified.** We give you the product and a few fixed constraints; every other technical decision is yours to make and to justify. We're grading how you reason under ambiguity — a confidently justified decision reads better than an unexplained one, even if we'd have chosen differently.

FIXED CONSTRAINTS

- Backend in **Python**, frontend in **TypeScript**.

- Market data from **Twelvedata**, with the key kept strictly server-side.
- Everything else — framework, storage, charts, auth, packaging — is open, and yours to defend.

WHAT TO BUILD

A single-page dashboard, gated behind login:

- **Authentication** — unauthenticated users cannot retrieve data.
- **Tickers** — a fixed universe of 5–10 symbols (e.g. AAPL, MSFT, NVDA, TSLA, AMZN), with a sensible default.
- **Interval** — daily required; intraday is a stretch goal.
- **Date range** — a sensible bounded default (e.g. last 3–6 months).
- **Chart** — closing price per ticker; candlesticks optional.
- **Summary panel** — % change, high/low, cumulative volume.

Surface volume somewhere, and handle **loading, error, and empty** states explicitly — distinguish a backend error from a genuine "no data" result.

DESIGN DECISIONS TO DOCUMENT

In place of a detailed spec, tell us how you approached the problems below — briefly, in your README, focusing on your reasoning and trade-offs.

1. **Data.** How did you store and structure the market data, and why is that the right fit? What would you reconsider at 100× the volume?
2. **Reliability.** The importer will be run repeatedly. How did you make sure that's always safe?
3. **Performance.** What keeps the dashboard responsive as the data grows, and where did you expect the first bottleneck?
4. **API.** How did you decide what the backend exposes and in what shape, and why that over the alternatives?
5. **Access control.** How did you restrict data to authenticated users, and how does the frontend handle that over a session?
6. **Frontend.** How did you keep the interface smooth and the code maintainable across a year of data for several tickers?

7. **Operations.** How would this run day to day in production?

HOW WE'LL EVALUATE

Weighted roughly evenly across backend and frontend. Beyond the product checklist, we're looking at the quality of your design reasoning, the clarity of your README, and your ability to talk us through your code afterwards.

OPTIONAL STRETCH GOALS

A complete, well-reasoned core beats any of these: intraday intervals · candlestick rendering · self-registration · one-command containerised setup · a derived "net-flow" metric from signed volume · or surprise us.

WHAT TO SUBMIT

A Git repository or zip archive with a **README** covering:

- setup & run for the backend, frontend, and importer (state Python/Node versions);
- the environment variables required;
- your answers to the design questions above;
- key trade-offs and assumptions; and
- what you'd do with more time.

If anything is unclear or you hit a blocker, reach out — though documenting a reasonable assumption and moving on is equally good signal.

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